CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

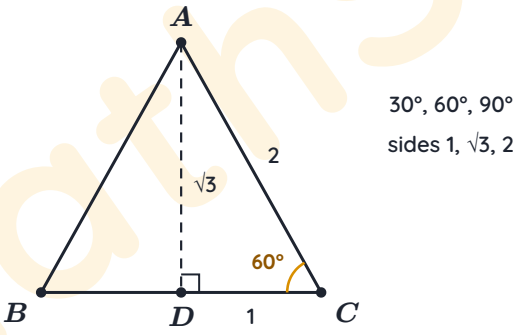
Chapter 23: Trigonometrical Ratios of Standard Angles

Topic-wise short notes with solved examples

The values of the trigonometrical ratios of 0°, 30°, 45°, 60° and 90°, how they are obtained, and their use in evaluating expressions and solving simple trigonometric equations.

1 Where the values come from

- $45°$: in an isosceles right triangle with equal sides 1, the hypotenuse is $\sqrt{2}$. So $\sin 45° = \cos 45° = \frac{1}{\sqrt{2}}$ and $\tan 45° = 1$.
- $30°$ **and** $60°$: in an equilateral triangle of side 2, an altitude makes two right triangles with angles $30°, 60°, 90°$ and sides $1, \sqrt{3}, 2$.
- So $\sin 30° = \frac{1}{2}$, $\cos 30° = \frac{\sqrt{3}}{2}$, $\sin 60° = \frac{\sqrt{3}}{2}$, $\cos 60° = \frac{1}{2}$.



Half of an equilateral triangle of side 2

2 Table of values

Angle	0°	30°	45°	60°	90°
sin	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
tan	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	not defined
cosec	not defined	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
sec	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	not defined
cot	not defined	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0

- **Memory aid for sine**: $\sqrt{\frac{0}{4}}, \sqrt{\frac{1}{4}}, \sqrt{\frac{2}{4}}, \sqrt{\frac{3}{4}}, \sqrt{\frac{4}{4}}$. The cosine row is the sine row reversed.
- As the angle increases from 0° to 90°: $\sin$ and $\tan$ **increase**; $\cos$ **decreases**.
- $\sin\theta = \cos\theta$ only at $\theta = 45^\circ$.

3 Evaluating expressions

- Replace each ratio by its value, then simplify. Remember $\sin^2 30^\circ$ means $(\sin 30^\circ)^2$.

Example 3.1 Evaluate $\sin 30^\circ \cos 60^\circ + \cos 30^\circ \sin 60^\circ$.

SOLUTION

$$\frac{1}{2} \times \frac{1}{2} + \frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2} = \frac{1}{4} + \frac{3}{4}$$

1

Example 3.2 Evaluate $2\tan^2 45^\circ + \cos^2 30^\circ - \sin^2 60^\circ$.

SOLUTION

$$2 + \frac{3}{4} - \frac{3}{4}$$

2

Example 3.3 Evaluate $\dfrac{\tan 60^\circ - \tan 30^\circ}{1 + \tan 60^\circ \tan 30^\circ}$.

SOLUTION

$$\frac{\sqrt{3} - \frac{1}{\sqrt{3}}}{1+1} = \frac{\frac{2}{\sqrt{3}}}{2}$$

$\dfrac{1}{\sqrt{3}}$, **which is** $\tan 30^\circ$

Example 3.4 Evaluate $4\sin^2 60^\circ + 3\tan^2 30^\circ - 8\sin 45^\circ \cos 45^\circ$.

SOLUTION

$$4 \times \frac{3}{4} + 3 \times \frac{1}{3} - 8 \times \frac{1}{2}$$

0

4 Verifying results

- Substitute the given angle on both sides and check that the values agree.
- **Warning**: $\sin(A+B)$ is **not** $\sin A + \sin B$, and $\sin 2A$ is **not** $2\sin A$.

Example 4.1 If $A = 30°$, verify that $\sin 2A = 2 \sin A \cos A$.

SOLUTION

LHS $= \sin 60° = \frac{\sqrt{3}}{2}$. RHS $= 2 \times \frac{1}{2} \times \frac{\sqrt{3}}{2}$.

Both equal $\frac{\sqrt{3}}{2}$

Example 4.2 If $A = 60°$ and $B = 30°$, verify that $\cos(A - B) = \cos A \cos B + \sin A \sin B$.

SOLUTION

LHS $= \cos 30° = \frac{\sqrt{3}}{2}$. RHS $= \frac{1}{2} \times \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} \times \frac{1}{2}$.

Both equal $\frac{\sqrt{3}}{2}$

Example 4.3 Show that $\sin 60° \neq 2 \sin 30°$.

SOLUTION

$\sin 60° = \frac{\sqrt{3}}{2}$, but $2 \sin 30° = 1$

They are different

5 Solving for an angle

- Match the given value with the table. In this chapter the angles lie between $0°$ and $90°$.
- If the angle is 2θ or $\theta + 15°$ and so on, find that angle first and then solve for θ.

Example 5.1 Find the acute angle θ if $2 \sin \theta = \sqrt{3}$.

SOLUTION

$\sin \theta = \frac{\sqrt{3}}{2}$

$\theta = 60°$

Example 5.2 Find θ if $\tan 3\theta = 1$ and 3θ is acute.

SOLUTION

$3\theta = 45°$

$\theta = 15°$

Example 5.3 Solve $2\cos^2\theta - 1 = 0$ for an acute angle.

SOLUTION

$\cos^2\theta = \frac{1}{2}$, so $\cos\theta = \frac{1}{\sqrt{2}}$

$\theta = 45^\circ$

Example 5.4 Find A and B if $\sin(A+B) = 1$ and $\cos(A-B) = \frac{\sqrt{3}}{2}$, where $A + B \leq 90^\circ$ and $A > B$.

SOLUTION

$A + B = 90^\circ$ and $A - B = 30^\circ$

$A = 60^\circ$, $B = 30^\circ$

Example 5.5 Solve $2\sin^2\theta - 3\sin\theta + 1 = 0$ for $0^\circ < \theta \leq 90^\circ$.

SOLUTION

$(2\sin\theta - 1)(\sin\theta - 1) = 0$, so $\sin\theta = \frac{1}{2}$ or $\sin\theta = 1$

$\theta = 30^\circ$ **or** $\theta = 90^\circ$

6 Finding sides

Example 6.1 In triangle ABC, $\angle B = 90^\circ$, $\angle A = 30^\circ$ and $AC = 12$ cm. Find BC and AB.

SOLUTION

$BC = AC\sin 30^\circ = 6$ cm and $AB = AC\cos 30^\circ = 6\sqrt{3}$ cm

6 cm and $6\sqrt{3}$ cm

7 Common mistakes

- Swapping the values of $\sin 30^\circ$ and $\sin 60^\circ$.
- Writing $\tan 90^\circ = 0$ or $\cot 0^\circ = 0$. They are not defined.
- Treating $\sin(A+B)$ as $\sin A + \sin B$.
- Reading $\sin^2 45^\circ$ as $\sin 90^\circ$.
- Giving θ when the equation was solved for 2θ or 3θ.
- Leaving a surd in the denominator when a rationalised answer is expected.